Backtracking

Monday, August 31, 2026 10:53 PM

15 Puzzle Problem

422

heuristic

current state

goal state

16 → 1 blank tiles

15 no.

1	4	2	3
13	6	7	8
5	10	11	
9	14	15	12

(a)

4x4 (initial state)

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

(b)

goal state

16! ways it can be arranged ≈ 21 trillion

Solvable config = $\frac{16!}{2}$ = 10.5 trillion

row-4	2	3	4
row-3 5	6	7	8
row-2 9	10	11	12
row-1 13	15	14	—

unsolvable

initial config

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 15, 14

inversion = 1

row = 1

sum = 2 → even ✗

→ odd ✓

1	2	3	4
row-3 5	6		8
9	10	7	11
13	14	15	12

solvable

initial config

1, 2, 3, 4, 5, 6, 8, 9, 10, 7, 11, 13, 14, 15, 12

inversions = 3 + 3 = 6

row = 3

sum = 9 odd

1	2	3	4
5	6	7	8
9	10	11	12
13	15	14	—

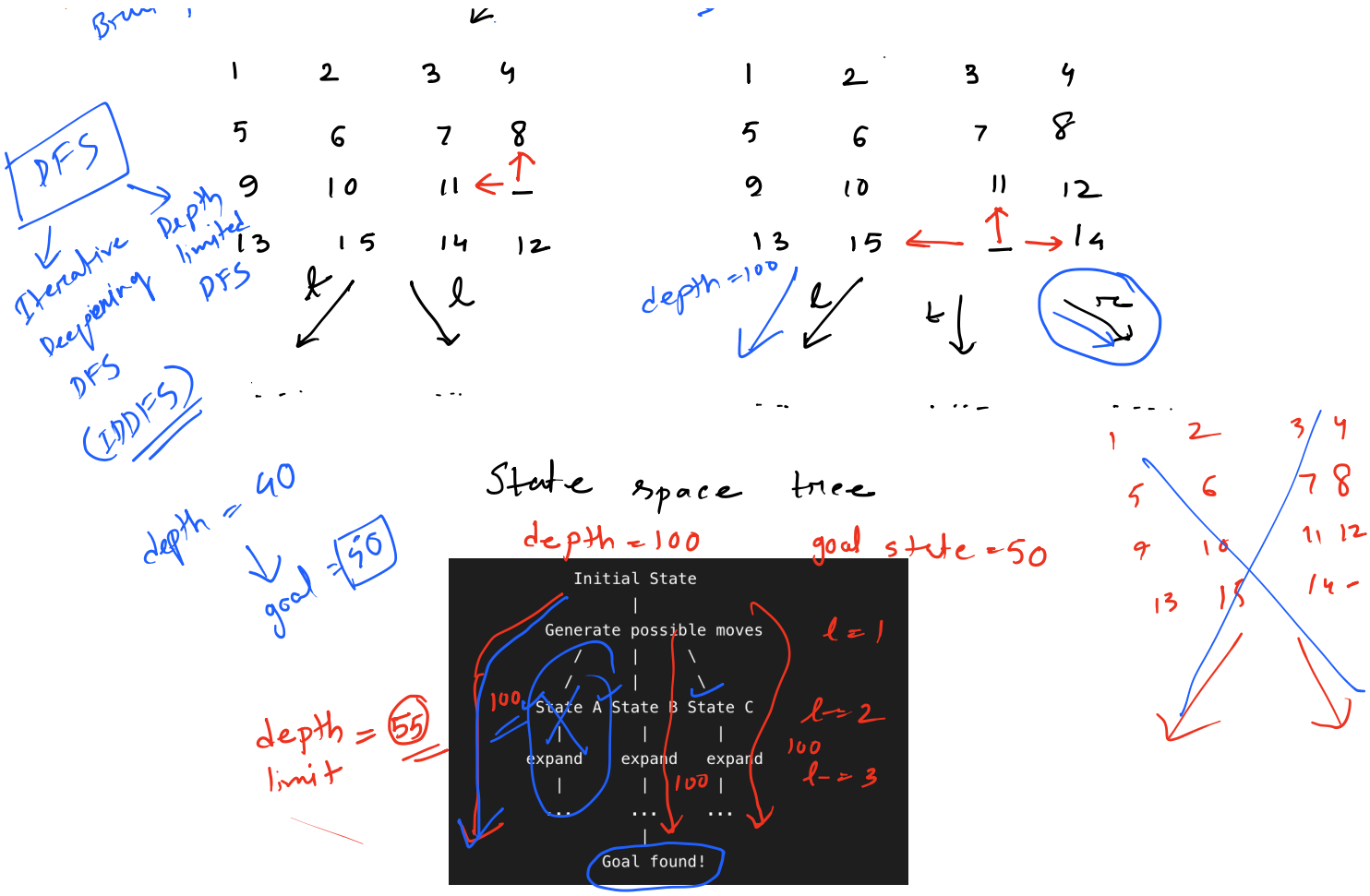
initial

top

left

Blind BFS/DFS → exponential time complexity

brute force


DFS
Iterative Deepening DFS
Depth limited DFS
(IDDFS)
depth = 40
goal = 40
depth = 55
limit
State space tree
depth = 100
goal state = 50
Initial State
Generate possible moves
State A
State B
State C
expand
expand
expand
100
100
100
l = 1
l = 2
l = 3
Goal found!
depth = 100